Additional Questions As Per New NCERT Text Book 2024-25 Edition

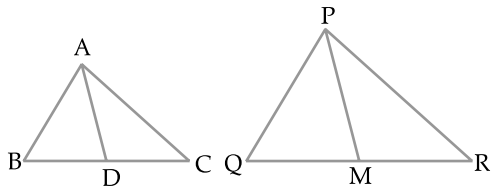
Chapter 7 : Triangles

NCERT CORNER

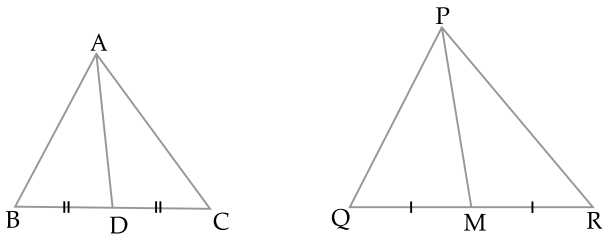
Textbook Questions

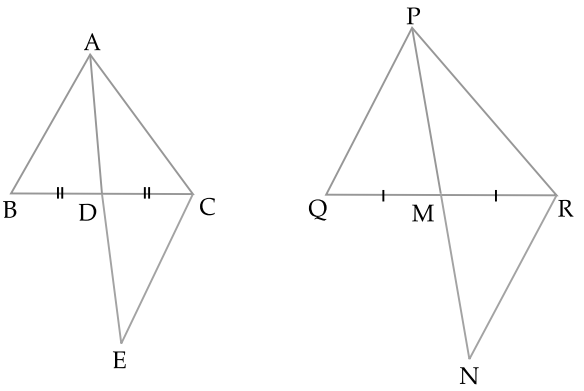
Exercise-6.3 (NCERT pg. no. 97) (For Oswaal One for All Book Page No. 180)

Q. 14. Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that Δ ABC ~ Δ PQR.



Sol.





Given: AD is the median of ΔABC, then

$$BD = CD = \frac{1}{2} BC \quad \ldots (1)$$

PM is the median of ΔPQR, then

$$QM = MR = \frac{1}{2} QR \quad \ldots (2)$$

And,

$$\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM} \quad \ldots (3)$$

Construction: Produce AD to E and join CE such that AD = DE
Similarly produce PM to N and join RN such that PM = MN.
To prove: $\Delta ABC \sim \Delta PQR$

Proof: Now, In Δ ABD and Δ DEC

BD = DC (AD is median)

∠BDA = ∠EDC (vertically opposite angles)

AD = DE (By construction)

∴ Δ ABD ~ Δ DEC (SAS Case)

So, AB = CE (CPCT) ….(i)

Similarly, Δ PQM ~ Δ MNR (SAS Case)

So, PQ = NR (CPCT) ...(ii)

As, $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{AD}{PM}$ (given)

And, From (i) and (ii) we get,

$$\frac{CE}{RN} = \frac{BC}{QR} = \frac{AD}{PM}$$

$$\Rightarrow \frac{CE}{RN} = \frac{BC}{QR} = \frac{2AD}{P2M}$$

$$\Rightarrow \frac{CE}{RN} = \frac{BC}{QR} = \frac{AE}{PN}$$

Thus, Δ ACE ~ Δ PNR (SSS Case)

∴ ∠CAE = ∠RPM (CPCT) ...(iii)

Similarly, Δ ABD ~ Δ PQM (SSS Case)

∴ ∠BAD = ∠QPM (CPCT) …(iv)

By Adding (iii) and (iv) we get,

∠BAC = ∠QPR ...(v)

Now, In, Δ ABC and Δ PQR

$$\frac{AB}{PQ} = \frac{AC}{PR}$$

∠BAC = ∠QPR (proved above)

∴ Δ ABC ~ Δ PQR (by SAS similarity).

Chapter 9 : Introduction to Trigonometry

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Textbook Questions

Exercise-8.3 (NCERT pg. no. 132) [(For Oswaal One for All Book Page No. 245/Q.4(x)]

Q. 1. $\left(\frac{1+\tan^2 A}{1+\cot^2 A}\right) = \left(\frac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A$

Sol. LHS= $\left(\frac{1+\tan^2 A}{1+\cot^2 A}\right)$ (∵ $1 + \tan^2 A = \sec^2 A$, and $1 + \cot^2 A = \text{cosec}^2 A$)

$$= \left(\frac{\sec^2 A}{\text{cosec}^2 A}\right)$$

$$= \left(\frac{\sin^2 A}{\cos^2 A}\right) = \tan^2 A$$ …(1)

Middle term: $\left(\frac{1-\tan A}{1-\cot A}\right)^2$

$$= \left(\frac{1-\tan A}{1-\frac{1}{\tan A}}\right)^2$$

$$= \left[\left(\frac{1-\tan A}{\tan A-1}\right)\times \tan A\right]^2$$

$$= \left(\frac{1-\tan A}{-(1-\tan A)}\right)^2 \times \tan^2 A$$

$= \tan^2 A = \text{LHS}.$

Thus RHS = MDS = LHS.

Hence proved.